PID vs. MRAC Control Techniques Applied to a Quadrotor's Attitude

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Abstract: Attitude is the key parameters for a Quadrotor. This paper designed a Proportional Integral Differential (PID) controller and a Model Reference Adaptive Control (MRAC) controller for a Quadrotor. At the end, a simulation was introduced. The simulation result told that MRAC controller had the ability to avoid the noise with the power such as 0.01.

Keywords: Quadrotor, PID, MRAC.

I. INTRODUCTION

Recently, the MEMS techniques make the development of unmanned aerial vehicle (UAV) possible[1]; and many scholars distribute themselves to the study of an unmanned aerial vehicle called Quadrotor. Quadrotor is a helicopter with four rotor wings which give the lift force to the helicopter. One example is show in Figure 1.The Quadrotor has the helicopter's characteristics such as vertical taking-off and vertical landing. The important characteristics of quadrotor are light and micro. Now Quadrotor is making an important role in rescue act, remote inspect task.

Erdinc Altug[2] studied the controller with visual feedback to quadrotor's attitude. Samir Bouadballah[3] gave the differences between PID controller and LQ controller in an indoor micro quadrotor. T. Madani and A. Benallegue[4] use Back-stepping controller to control quadrotor's attitude and position. Mehmet Onder Efe[5] use sliding models to design a robust controller to quadrotor. This study is to explore the differences between PID controller and MRAC controller to quadrotor's attitude.

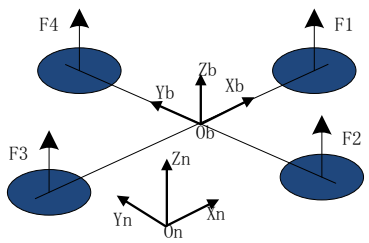


Figure 1 Configure of Quadrotor

In section 2 the quadrotor's dynamics and model is showed. The PID controller and MRAC controller are displayed in section 3.In the last section a simulation is described. The differences between PID and MRAC are described in section 4.

II. QUADROTOR MODEL

Quadrotor which has four fix angle rotors to make the lift force is different from common helicopter. Quadrotor has two flying styles one called X style and the other called + style.

This study is to describe the + style's attitude controller. On + style, rotor 1 and 3 determine the pitch moment; rotor 2 and 4 determine the roll moment. The four rotors determine the yaw moment.

2.1 Dynamics of Quadrotor

On the assumption that the frame which fixed on the quadrotor's body is E(xb, yb, zb); axis xb is on the line which is fixed by rotor 1 and 3; axis yb is on the line which is determined by rotor 2 and 4; and the zb axis is perpendicular to the plane which is configured by axis xb and yb. If the speed of rotor 1 is great than the speed of rotor 3, the pitch angle is positive. The roll angle is positive when the speed of rotor 2 is great than rotor 4.If the sum of rotor 1's and rotor 3's speed is great than the sum of rotor 2 and rotor 4, yaw angle is positive.

The body coordinate frame and navigation coordinate frame are described in figure 1. This study assumes these:

1. Quadrotor is rigid body, the center of gravity is the same as geometric center.
2. The rotors of quadrotor are fixed and unreformed.

On the assumption in up contents, the equation of motion of quadrotor is described as next:

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$$\begin{aligned}\dot{\varepsilon} &= v \\ m\dot{v} &= RF_b \\ \dot{R} &= RS(\Omega) \\ J\dot{\Omega} &= -\Omega \times J\Omega + F_f\end{aligned} \tag{1}$$

ε is the position vector; v is the velocity vector; the rotation array is R; m is the mass of quadrotor; S(Ω) is an anti-symmetric matrix for Ω which is the angular velocity vector. J is the moment of inertia. F_f is the input torque on the body frame.

On the assumption that the quadrotor is rigid body, the rotation array R can be described like equation 2.

$$\mathrm{R}=\begin{bmatrix} C\psi C\theta & S\psi C\theta & -S\theta \\ C\psi S\theta S\phi - S\psi C\phi & S\psi S\theta S\phi + C\psi C\phi & C\theta S\phi \\ C\psi S\theta C\phi + S\psi S\phi & S\psi S\theta C\phi - C\psi S\phi & C\theta C\phi \end{bmatrix} \tag{2}$$

In equation 2, C(.) is the same as cos(.); S(.) is the same as sin(.).

The moment of inertia is described as matrix like equation 3.

$$J = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \tag{3}$$

This study assume that F1, F2, F3, F4 is the force output of motor 1, motor 2, motor 3 and motor 4. The roll moment is U2; U3 is the pitch moment; the yaw moment is U4. Equation 4 is described the relation of the moments and the forces.

$$\begin{aligned} U_2 &= l(F_2 - F_4) \\ U_3 &= l(F_1 - F_3) \\ U_4 &= cl(F_1 - F_2 + F_3 - F_4) \end{aligned} \tag{4}$$

In Equation 4，l is the arm of force, c is a const.

According to equation 1 and equation 4:

$$\begin{aligned} I_{xx}\ddot{\phi} &= \dot{\theta}\dot{\varphi}(I_{yy} - I_{zz}) + U_2 \\ I_{yy}\ddot{\theta} &= \dot{\phi}\dot{\varphi}(I_{zz} - I_{xx}) + U_3 \\ I_{zz}\ddot{\varphi} &= \dot{\phi}\dot{\theta}(I_{xx} - I_{yy}) + U_4 \end{aligned} \tag{5}$$

2.2 problems

Quadrotor has two controllers, one is to attitude and the other one is to navigation[6]. The navigation controller often contains the man control, also can use another navigation control technique.

According to equation 5, the model of quadrotor's attitude is nonlinear. In order to design an controller to quadrotor via linear theories, there are many methods which can make the nonlinear system be linear such as feedback linearization[7], approach linearization. In order to design controller easily, approach linearization is the linearization method of this study, so assume that

$$\begin{aligned} &-25° < \phi < +25° \\ &-25° < \theta < +25° \\ &-180° < \varphi < +180° \\ &-10°/s < \dot{\phi} < +10°/s \\ &-10°/s < \dot{\theta} < +10°/s \\ &-10°/s < \dot{\varphi} < +10°/s \end{aligned} \tag{6}$$

III. DESIGN OF PID AND MRAC CONTROLLER

3.1 linear quadrotor' s model

On the Equation 6 described control needs, this study makes the quadrotor's attitude model be linear as equation 7.

$$\begin{aligned} I_{xx}\ddot{\phi} &= U_2 + \varepsilon_2 \\ I_{yy}\ddot{\theta} &= U_3 + \varepsilon_3 \\ I_{zz}\ddot{\varphi} &= U_4 + \varepsilon_4 \end{aligned} \tag{7}$$

In equation 7, ε2 replaces the noise of roll; ε3 is noise of pitch; ε4 replaces the noise of yaw.

Assuming the states as equation 8; the study can describe the quadrotor in states space style as equation 9.

$$\begin{aligned} x_1 &= \phi \\ x_2 &= \dot{x}_1 = \dot{\phi} \\ x_3 &= \theta \\ x_4 &= \dot{x}_3 = \dot{\theta} \\ x_5 &= \varphi \\ x_6 &= \dot{x}_5 = \dot{\varphi} \end{aligned} \tag{8}$$

Let X=[x1, x2, x3, x4, x5, x6], then

$$\dot{X} = f(X,U) = \begin{bmatrix} x_2 \\ \frac{U2}{I_{xx}} \\ x_4 \\ \frac{U_3}{I_{yy}} \\ x_6 \\ \frac{U_4}{I_{zz}} \end{bmatrix} + \varepsilon \qquad (9)$$

In equation 9, $\varepsilon=[0, \varepsilon_2/I_{xx}, 0, \varepsilon_3/I_{yy}, 0, \varepsilon_4/I_{zz}]$.

3.2 Design of PID controller

The Laplace Transform of equation 7 like equation 10.

$$\begin{aligned} I_{xx}s^2\phi(s) &= U_2(s) + \varepsilon_2(s) \\ I_{yy}s^2\theta(s) &= U_3(s) + \varepsilon_3(s) \\ I_{zz}s\varphi(s) &= U_4(s) + \varepsilon_4(s) \end{aligned} \qquad (10)$$

Equation 10 displays that the pole points of quadrotor's pitch roll and yaw channel are zero; so these three channels are not stability.

In order to come true stable pitch channel, roll channel and yaw channel; this study use PD controllers like equation 11 describing.

$$\begin{aligned} u_2 &= k_{p\phi}(\phi_{cmd} - \phi_{fdk}) + k_{d\phi}\frac{d(\phi_{cmd} - \phi_{fdk})}{dt} \\ u_3 &= k_{p\theta}(\theta_{cmd} - \theta_{fdk}) + k_{d\theta}\frac{d(\theta_{cmd} - \theta_{fdk})}{dt} \\ u_4 &= k_{p\varphi}(\varphi_{cmd} - \varphi_{fdk}) + k_{d\varphi}\frac{d(\varphi_{cmd} - \varphi_{fdk})}{dt} \end{aligned} \qquad (11)$$

In equation 11, u_2 is the output of roll PD controller; the output of pitch PD controller is u_3; u_4 is the output of yaw PD controller. The roll proportional gain is $k_{p\Phi}$, differential gain $k_{d\Phi}$.The pitch proportional gain is $k_{p\theta}$, differential gain is $k_{d\theta}$. The yaw proportional gain is $k_{p\psi}$, differential gain is $k_{d\psi}$.The roll pitch and yaw reference inputs are Φ_{cmd},θ_{cmd}, ψ_{cmd}. And the feedbacks of roll pitch and yaw channel are Φ_{fdk}, θ_{fdk}, ψ_{fdk}.

3.3 Design of MRAC controller

MRAC controller design is a method which uses a model to design an adaptive controller[8]. MRAC controller can make the states or outputs of object approach to the states or outputs of the model. This study select three model described in equation 12, equation 13 and equation 14 for pitch roll and yaw channel.

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -10 & -10 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_2 \quad (12)$$

$$\begin{bmatrix} \frac{dx_3}{dt} \\ \frac{dx_4}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -10 & -10 \end{bmatrix}\begin{bmatrix} x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_3 \quad (13)$$

$$\begin{bmatrix} \frac{dx_5}{dt} \\ \frac{dx_6}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -10 & -10 \end{bmatrix}\begin{bmatrix} x_5 \\ x_6 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_4 \quad (14)$$

According to Lyapunov's law and states feedback, design a MRAC controller. Select Lyapunov function as equation 15.

$$V = \frac{1}{2}e^T P e + \mathrm{tr}(\emptyset^T T_1^{-1}\emptyset + \varphi^T T_2^{-1}\varphi) \qquad (15)$$

In equation 15, $\emptyset = \bar{F} - F$, $\varphi = \bar{K} - K$; $\bar{F}$ is the desire feedback gain, F is the feedback gain. $\bar{K}$ is the desire feedback gain, K is the feedback gain.

According to Lyapunov's law, get the feedback gain F and feed-forward gain K.

$$F(t) = \int_0^t T_1(B_m\bar{K}^{-1})^T P e x_p^T d\tau + F(0) \qquad (16)$$

$$K(t) = \int_0^t T_2(B_m\bar{K}^{-1})^T P e r^T d\tau + K(0) \qquad (17)$$

In equation 16 and equation 17, P, T1, T2 are positive definite symmetric matrix; $B_m\bar{K}^{-1} = B_p$; $e = x_m - x_p$.

IV. Simulations and Conclusion

These simulations use the parameters from the reference paper 3. Let $I_{xx} = I_{yy} = 0.0075N/m^2$, $I_{zz} = 0.0013N/m^2$. And let $k_{p\emptyset} = k_{p\theta} = 5$, $k_{d\emptyset} = k_{d\theta} = 1$; $k_{p\varphi} = 2$; $k_{d\varphi} = 1.8$; $H_\emptyset = H_\theta = 1$, $H_\varphi = 200$. Let the noise power be 0.01. The simulation which use PD controller to quadrotor's attitude control is displayed in figure 2 , figure 3, figure 4.

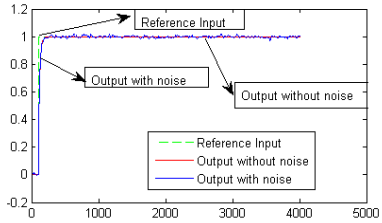


Figure 2 Roll channel simulation of PD

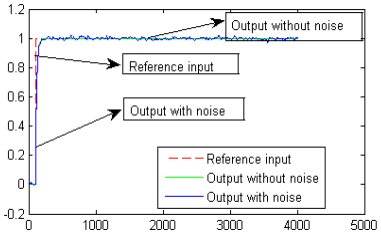


Figure 3 Pitch channel simulation of PD

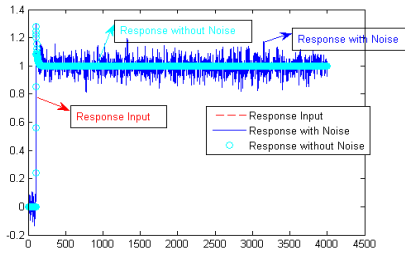


Figure 4 Yaw channel simulation of PD

According to equation 11, 12, 13, 14, 15,16,17, let P=[3,1; 1,1], T_1=T_2=1 and F(0)=K(0)=0; let the noise's power be 0.01. The simulations which use MRAC controller are showed like figure 5, figure 6 and figure 7.

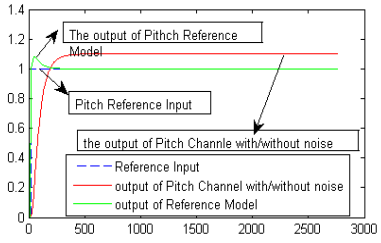


Figure 5 Pitch channel simulation of MRAC

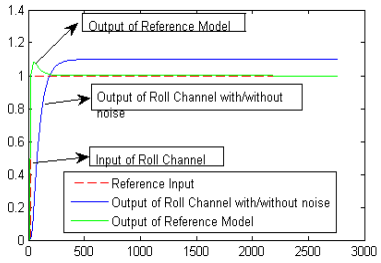


Figure 6 Roll channel simulation of MRAC

According the simulations, this study has this conclusion: PID controller is very easy to design but if there are some noises such as a noise whose power is 0.01, PID controller can't avoid the influence of the noise. MRAC controller is

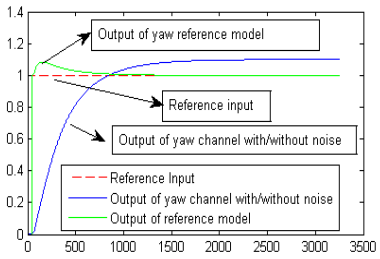


Figure 7 Yaw channel simulation of MRAC

difficult but this method has robust. MRAC is good method when controller designer want the controller can avoid the influence of noise from the object insert or exterior.

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